

Ethan Ashley

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PORTFOLIO: <https://mechethan.github.io/>

Education

NC State University

Master of Science in Mechanical Engineering — *Expected Fall 2027*

Focus: Mechanical and Electrical Design

Auburn University

Bachelor of Science in Mechanical Engineering — *May 2025*

GPA: 3.2

Technical Skills

CAD & CAM: SolidWorks (Advanced), Inventor, Mastercam, GD&T

Simulation & Analysis: Adams View (Dynamic Systems), MATLAB, Python (basic)

Manufacturing: Composite manufacturing, 3D printing, CNC, milling machines, lathes

Electrical / Mechatronics: Circuit design fundamentals, microcontroller integration

Other: Microsoft Excel & PowerPoint, Technical documentation, Data analysis

Engineering Experience

Lean Engineering Intern

GKN Aerospace | Tallahassee, AL — Composites | *Aug 2025 – Dec 2025*

- Supported composite manufacturing and quality improvement efforts for **GENx engine fan case abradable panels**, applying Lean principles in a high-rate aerospace production environment.
- Led **quality review and defect analysis** through destructive testing, visual inspection, and data-driven evaluation, supporting **MRB and RCCA** decision-making.
- Approved approximately **360 previously scrapped composite parts for production**, recovering **~\$2.5M in material value** and directly contributing to scrap reduction initiatives.
- Verified and dispositioned an additional **~360 parts as scrap**, ensuring compliance with aerospace quality standards and preventing nonconforming material from reaching production.
- Utilized **Excel-based data analysis** to track defect trends, evaluate process capability, and support continuous improvement initiatives.
- Created and delivered **400+ PowerPoint technical reports** to communicate defects, trends, and recommendations to operators, engineers, and management.

- Collaborated cross-functionally with **manufacturing, quality, and design engineers** to support daily production and process improvement efforts.
- Served as the **initial GKN employee supporting the Abradable MRB process**, coordinating with engineers from multiple global GKN sites.
- Gained hands-on exposure to **Lean Manufacturing, Six Sigma, MRB, RCCA, GD&T**, and the **TipQA quality management system**.

Additive Manufacturing — Senior Design Project

Auburn University | *Aug 2024 – May 2025*

- Collaborated with Auburn's **National Center for Additive Manufacturing Excellence (NCAME)** and the **U.S. Army Aviation and Missile Center**.
- Researched metal additive manufacturing processes and material capabilities.
- Designed and modeled a metal additive benchmark artifact using **SolidWorks and Mastercam** for FormAlloy's L5 LP-DED printer.

Rocketry Association & Small Satellite Program

Auburn University | *Aug 2021 – May 2023*

- Designed, built, and tested payload components for **NASA-sponsored competitions**.
- Conducted mechanical systems integration, CAD modeling, and iterative prototype testing.
- Participated in Small Satellite Program training for constructing government-funded scientific satellites.

Leadership & Activities

- **Eagle Scout**
- Auburn University Marching Band & Basketball Pep Band — 4 years
- Kappa Kappa Psi National Honorary Band Fraternity — 4 years
- American Society of Mechanical Engineers (ASME) Member
- International Learning Participant — Honors College Study Abroad (London/Paris/Italy)
- Volunteer, Auburn Honors Congress

Certifications

- ITAR Certification (2025)
- Cybersecurity Awareness Training (2024)